

Day 5: Strings in Python (Detailed Notes for Data Analysis Students)

What is a String?

A string is a sequence of characters enclosed in quotes.

Examples:

```
name = "Umesh"  
city = 'Mumbai'
```

Strings can contain:

- Letters
- Numbers
- Symbols
- Spaces

Why Are Strings Important?

In Data Analysis, most raw data is stored as text.

Examples:

Data Type	Example
Customer Name	Rahul Sharma
Email	rahul@gmail.com
City	Mumbai

Product Name	Laptop
Phone Number	9876543210

Before analysis, text data often needs cleaning and formatting.

Creating Strings

```
name = "Umesh"  
  
print(name)
```

Output:

```
Umesh
```

String Length

Use `len()`.

```
name = "Python"  
  
print(len(name))
```

Output:

```
6
```

String Indexing

Each character has a position.

```
P Y T H O N
0 1 2 3 4 5
```

Example:

```
text = "Python"

print(text[0])
```

Output:

```
P
```

Negative Indexing

```
P Y T H O N
-6 -5 -4 -3 -2 -1
```

```
text = "Python"

print(text[-1])
```

Output:

```
N
```

String Slicing

Syntax:

```
string[start:end]
```

Example:

```
text = "Python"  
  
print(text[0:3])
```

Output:

```
Pyt
```

Examples

```
text = "DataAnalysis"  
  
print(text[0:4])
```

Output:

```
Data
```

```
text = "Python"  
  
print(text[2:6])
```

Output:

```
thon
```

String Concatenation

Joining strings.

```
fname = "Umesh"  
lname = "Chauhan"  
  
print(fname + " " + lname)
```

Output:

Umesh Chauhan

String Repetition

```
print("Python " * 3)
```

Output:

```
Python Python Python
```

Convert to Upper Case

```
name = "umesh"  
  
print(name.upper())
```

Output:

```
UMESH
```

Convert to Lower Case

```
name = "UMESH"  
  
print(name.lower())
```

Output:

```
umesh
```

Capitalize First Letter

```
name = "umesh"  
  
print(name.capitalize())
```

Output:

```
Umesh
```

Title Case

```
name = "umesh chauhan"  
  
print(name.title())
```

Output:

```
Umesh Chauhan
```

Remove Extra Spaces

```
name = "  Umesh  "  
  
print(name.strip())
```

Output:

```
Umesh
```

Replace Text

```
text = "I Love Excel"  
  
print(text.replace("Excel", "Python"))
```

Output:

```
I Love Python
```

Find Text

```
text = "Data Analysis"

print(text.find("Analysis"))
```

Output:

```
5
```

Check Starting Text

```
email = "admin@gmail.com"

print(email.startswith("admin"))
```

Output:

```
True
```

Check Ending Text

```
file = "sales.xlsx"

print(file.endswith(".xlsx"))
```

Output:

```
True
```

Split Method

Converts string into list.

```
text = "Python,SQL,Excel"  
  
print(text.split(","))
```

Output:

```
['Python', 'SQL', 'Excel']
```

Join Method

Combines list into string.

```
skills = ['Python','SQL','Excel']  
  
print(",".join(skills))
```

Output:

```
Python,SQL,Excel
```

Count Characters

```
text = "Python"  
  
print(text.count("o"))
```

Output:

```
1
```

String Formatting

Method 1

```
name = "Rahul"  
  
print("Welcome", name)
```

Method 2 (Recommended)

```
name = "Rahul"  
salary = 50000  
  
print(f"{name} earns {salary}")
```

Output:

```
Rahul earns 50000
```

Real Data Analyst Examples

Example 1: Clean Customer Names

```
customer = " rahul sharma "  
  
customer = customer.strip().title()  
  
print(customer)
```

Output:

```
Rahul Sharma
```

Example 2: Email Validation

```
email = "abc@gmail.com"

if email.endswith("@gmail.com"):
    print("Valid Gmail")
```

Example 3: Product Name Cleaning

```
product = "Laptop "

product = product.strip()

print(product)
```

Mini Project 1: Full Name Generator

```
fname = input("Enter First Name: ")
lname = input("Enter Last Name: ")

fullname = fname.title() + " " + lname.title()

print(fullname)
```

Mini Project 2: Email Checker

```
email = input("Enter Email: ")

if email.endswith("@gmail.com"):
    print("Valid Gmail")

else:
    print("Invalid Gmail")
```

Mini Project 3: Word Counter

```
text = input("Enter Text: ")

print("Length =", len(text))
```

Mini Project 4: Data Cleaning Tool

```
name = input("Enter Name: ")

clean_name = name.strip().title()

print(clean_name)
```

Practice Questions

Basic

1. Create a string.
 2. Print a string.
 3. Find length of a string.
 4. Print first character.
 5. Print last character.
-

Indexing & Slicing

6. Print first 3 characters.
 7. Print last 4 characters.
 8. Extract city from a string.
 9. Reverse a string using slicing.
 10. Print alternate characters.
-

Methods

11. Convert to uppercase.
 12. Convert to lowercase.
 13. Capitalize text.
 14. Convert to title case.
 15. Remove spaces.
-

Replace & Find

16. Replace Python with SQL.
 17. Replace Mumbai with Pune.
 18. Find word position.
 19. Count occurrences of a character.
 20. Check if text exists.
-

Split & Join

21. Split comma-separated values.
 22. Split names.
 23. Join city names.
 24. Join product names.
 25. Join skills.
-

Data Cleaning

26. Remove spaces from names.
 27. Convert names to title case.
 28. Validate email IDs.
 29. Validate file extensions.
 30. Clean customer records.
-

Business Problems

31. Format employee names.
 32. Generate employee IDs.
 33. Validate email addresses.
 34. Extract domain names.
 35. Create sales report headers.
 36. Count customer names.
 37. Clean city names.
 38. Remove duplicate spaces.
 39. Format product names.
 40. Create invoice messages.
-

Assignment (Data Analyst Level)

Create a Customer Data Cleaning Program:

Input:

```
Customer Name : "  rahul sharma "  
Email : "RAHUL@GMAIL.COM"
```

Output:

```
Customer Name : Rahul Sharma  
Email : rahul@gmail.com  
Email Valid : True
```

Use:

- strip()
 - title()
 - lower()
 - endswith()
-

Interview Questions

1. What is a string?
2. What is indexing?
3. What is slicing?
4. Difference between upper() and lower()?
5. What is strip()?
6. What is split()?
7. What is join()?
8. What is replace()?
9. How are strings used in data cleaning?
10. Give a real-life example of string manipulation.

Learning Outcome

After Day 5, you should be able to:

- ✓ Create and manipulate strings
- ✓ Use indexing and slicing
- ✓ Apply string methods
- ✓ Clean text data
- ✓ Validate emails and file names
- ✓ Perform basic data-cleaning tasks used in Data Analysis